

WORK SHEET - 12

ELLIPSE & HYPERBOLA

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.8 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 If PQ is a double ordinate of the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  such that OPQ is an equilateral triangle, O being the centre of the hyperbola, then the eccentricity  $e$  of the hyperbola satisfies  
(A)  $1 < e < \frac{2}{\sqrt{3}}$  (B)  $e = \frac{2}{\sqrt{3}}$  (C)  $e = \frac{\sqrt{3}}{2}$  (D)  $e > \frac{2}{\sqrt{3}}$
- Q.2 If the ellipse  $4x^2 + 9y^2 = 36$  and the hyperbola  $\alpha^2 x^2 - y^2 = 4$  intersects at right angles, then the equation of circle passing through the points of intersection of two conics is  
(A)  $x^2 + y^2 = 4$  (B)  $x^2 + y^2 = 1$  (C)  $x^2 + y^2 = 3$  (D)  $x^2 + y^2 = 5$
- Q.3 A series of ellipses  $E_1, E_2, \dots, E_n$  with centre origin are drawn such that  $E_n$  touches  $E_{n-1}$  at the extremities of the major axis of  $E_{n-1}$  and the foci of  $E_n$  coincide with the extremities of minor axis of  $E_{n-1}$ . If eccentricity of the ellipses is independent of  $n$ , then the value of eccentricity is  
(A)  $\frac{\sqrt{5}}{3}$  (B)  $\frac{\sqrt{5}+1}{4}$  (C)  $\frac{\sqrt{5}-1}{2}$  (D)  $\frac{\sqrt{5}-1}{3}$
- Q.4 If  $P(\theta)$  and  $Q\left(\frac{\pi}{2} + \theta\right)$  are two points on the ellipse  $\frac{x^2}{9} + \frac{y^2}{4} = 1$ , then the locus of midpoint of PQ is  
(A)  $\frac{x^2}{9} + \frac{y^2}{4} = \frac{1}{2}$  (B)  $\frac{x^2}{9} + \frac{y^2}{4} = 2$  (C)  $\frac{x^2}{9} + \frac{y^2}{4} = 4$  (D)  $\frac{x^2}{9} + \frac{y^2}{4} = 8$
- Q.5 If a hyperbola whose foci are  $(-2, 4)$  and  $(4, 6)$  touches  $y$ -axis, then equation of hyperbola is  
(A)  $\frac{(3x+y-8)^2}{2} - \frac{(x-3y+14)^2}{8} = 1$  (B)  $\frac{(x-3y+14)^2}{2} - \frac{(3x+y-8)^2}{8} = 20$   
(C)  $\frac{(x+3y-7)^2}{2} - \frac{(x-3y+8)^2}{8} = 1$  (D)  $\frac{(3x+y-8)^2}{1} - \frac{(x-3y+14)^2}{4} = 20$
- Q.6 If the angle between ellipse  $x^2 + 4y^2 = 4$  and circle  $x^2 + y^2 - 2 = 0$  is  $\theta$ , then  $\tan \theta$  is equal to  
(A)  $\frac{1}{2}$  (B)  $\frac{1}{\sqrt{2}}$  (C)  $\frac{1}{2\sqrt{2}}$  (D)  $\frac{2}{3}$
- Q.7 The locus of centre of a variable circle touching the circles of radius  $r_1$  and  $r_2$  externally, which also touch each other externally, is a conic of the eccentricity  $e$ . If  $\frac{r_1}{r_2} = 3 + 2\sqrt{2}$ , then  $e^2$  is  
(A) 2 (B) 3 (C) 4 (D) 5

- Q.8 If the straight line  $4ax + 3by = 24$  is a normal to the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  ( $a > b$ ) then  $(a^2 - b^2)$  is equal to
- (A) 6 (B) 8 (C) 10 (D) 12

**[PARAGRAPH TYPE]**

**Q.9 to Q.14** has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

**Paragraph for question nos. 9 & 10**

Let a hyperbola having transverse axis and conjugate axis parallel to x-axis and y-axis respectively. Its two foci S and S' are (2, 1), (10, 1) and a straight line  $x + y - 9 = 0$  is tangent to this hyperbola at point P.

- Q.9 The equation of director circle of hyperbola is
- (A)  $x^2 + y^2 - 12x - 2y - 12 = 0$  (B)  $x^2 + y^2 - 12x - 2y + 12 = 0$   
 (C)  $x^2 + y^2 - 12x - 2y + 33 = 0$  (D)  $x^2 + y^2 - 12x - 2y - 33 = 0$

- Q.10 The length of latus rectum of hyperbola is

- (A)  $\frac{5\sqrt{10}}{6}$  (B)  $\frac{6}{5}\sqrt{10}$  (C)  $\frac{5\sqrt{5}}{6}$  (D)  $\frac{6}{5}\sqrt{14}$

**Paragraph for question nos. 11 & 12**

An ellipse has semi-major axis of length 2 and semi-minor axis of length 1. It slides between the co-ordinate axes in the first quadrant while maintaining contact with both x-axis and y-axis.

- Q.11 The locus of the centre of ellipse is
- (A)  $x^2 + y^2 = 3$  (B)  $x^2 + y^2 = 5$   
 (C)  $(x - 2)^2 + (y - 1)^2 = 5$  (D)  $(x - 2)^2 + (y - 1)^2 = 3$

- Q.12 The locus of the foci of the ellipse is

- (A)  $x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} = 16$  (B)  $x^2 + y^2 + \frac{1}{x^2} - \frac{1}{y^2} = 2\sqrt{3} + 4$   
 (C)  $x^2 + y^2 - \frac{1}{x^2} - \frac{1}{y^2} = 16$  (D)  $x^2 + y^2 - \frac{1}{x^2} + \frac{1}{y^2} = 2\sqrt{3} + 4$

**Paragraph for question nos. 13 & 14**

Let the foci of the hyperbola  $\frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$  be the vertices of the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  and the foci of the ellipse be the vertices of the hyperbola. Let the eccentricities of the ellipse and hyperbola be  $e_1$  and  $e_2$  respectively. Given  $b = 1$  and  $e_2 = 2$ .

- Q.13 The radius of director circle of the ellipse is equal to

- (A)  $\frac{2}{\sqrt{3}}$  (B)  $\sqrt{\frac{7}{3}}$  (C)  $\frac{\sqrt{3}}{2}$  (D)  $\sqrt{\frac{3}{7}}$

- Q.14 The length of the latus-rectum of the hyperbola is equal to

- (A)  $\frac{1}{\sqrt{3}}$  (B)  $\frac{2}{\sqrt{3}}$  (C)  $\sqrt{3}$  (D)  $2\sqrt{3}$



### [MULTIPLE CORRECT CHOICE TYPE]

**Q.15 to Q.17** has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.15 Let the straight line passing through  $T(\alpha, 2)$  meets the ellipse  $4x^2 + 9y^2 = 36$  at A and D and meets the coordinate axes at B and C such that TA, TB, TC, TD (taken in that order) are in G.P., then possible values of  $\alpha$  can be  
(A) -7 (B) 5 (C) 8 (D) 10
- Q.16 The differential equation  $\frac{dx}{dy} = \frac{3y}{2x}$  represents a family of hyperbolas (except when it represents a pair of lines) with eccentricity  
(A)  $\sqrt{\frac{5}{3}}$  (B)  $\sqrt{\frac{10}{3}}$  (C)  $\sqrt{\frac{5}{2}}$  (D)  $\sqrt{\frac{10}{7}}$
- Q.17 The locus of incentre of triangle  $PF_1F_2$  where P is a variable point on ellipse  $x^2 + 2y^2 = 4$  with  $F_1, F_2$  as foci is a conic whose  
(A) equation of auxiliary circle is  $x^2 + y^2 = 2$   
(B) eccentricity equals  $\sqrt{2(\sqrt{2} - 1)}$   
(C) equation of auxiliary circle is  $x^2 + y^2 = 4$   
(D) eccentricity equals  $\sqrt{2(\sqrt{2} + 1)}$

### [INTEGER TYPE]

**Q.18 to Q.19** are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

- Q.18 If from every point on the tangent to the parabola  $y^2 = 12x$  at its upper end of latus-rectum, pair of tangents are drawn to the hyperbola  $\frac{x^2}{4} - \frac{y^2}{1} = 1$ , then locus of mid-points of their corresponding chord of contact is  $3x^2 - ay^2 - bx - 4y = 0$ . Find the value of  $(a + b)$ .
- Q.19 From a point P, tangents are drawn to the curve  $\frac{x^2}{2} - y^2 = 1$ . If the chord of contact is a normal chord, then the locus of P is the curve  $\frac{8}{x^2} - \frac{1}{y^2} = \lambda$ , where  $\lambda \in \mathbb{N}$ . Find  $\lambda$ .

### ANSWER KEY

|      |    |      |    |      |     |      |     |      |     |
|------|----|------|----|------|-----|------|-----|------|-----|
| Q.1  | D  | Q.2  | D  | Q.3  | C   | Q.4  | A   | Q.5  | D   |
| Q.6  | B  | Q.7  | A  | Q.8  | C   | Q.9  | C   | Q.10 | B   |
| Q.11 | B  | Q.12 | A  | Q.13 | B   | Q.14 | D   | Q.15 | ACD |
| Q.16 | AC | Q.17 | AB | Q.18 | [8] | Q.19 | [9] |      |     |

